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Research note: The validity and reliability of a novel wearable sensor in measuring light exposure in an indoor and outdoor environment

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Light exposure has been measured previously using wrist-worn devices, but they may not reflect retinal light accurately. Few studies have evaluated the validity and reliability of light sensors placed near the eyes. This study aims to compare a wearable 'light button' to a factory-calibrated digital light meter (DLM) and evaluate the light button's inter-device reliability. Two light buttons and a DLM were tested side-by-side indoors and outdoors for 8 h. Lux values were compared using typical error of estimate (TEE), mean differences, 95% levels of agreement, Pearson correlation coefficient (r) and coefficient of variation (CV%). Inter-device reliability was assessed using TEE, mean differences and intraclass correlation coefficients (ICCs). Indoor validity analysis resulted in a TEE of 71.0 lx, mean difference was 24.7 lx, a strong correlation ($r = 0.96$) and CV of 11.5%. Outdoors, TEE was 45.3 lx, mean difference was 82.3 lx, with a strong correlation ($r = 1.00$) and CV of 1.8%. Inter-device reliability indoors and outdoors had TEEs of 10.7 lx and 33.5 lx, mean differences of 67.1 lx and 45.3 lx and strong ICCs of 0.90 and 0.98, respectively. The light buttons are valid and reliable for monitoring light exposure indoors and outdoors. Further validation is required in free-living conditions.

1. Introduction

The light-dark pattern is crucial for human circadian health and is required to synchronise the day-night cycle over 24 h[1] to regulate both the homeostatic drive (process S), and circadian drive (process C) of sleep regulation in humans.[2] Process S depicts sleep drive, whereas process C models sleep pressure. Light directly affects process C; however, previous studies have suggested that light also impacts process S indirectly through process C.[3] In recent years, there has been a surge in the number of wearable devices used in sleep, health, fitness and research.[4–6] Wearable light sensors may provide valuable information regarding 24-h light exposure in humans and how this may relate to a myriad of factors, including sleep, mood, well-being and performance. Until recently, monitoring light exposure in free-living conditions throughout the day has been difficult. New commercially available technologies that allow humans to wear clip-on light sensors show promise, though external validation studies are currently lacking.

Light exposure has historically been measured using wrist-worn devices in humans over prolonged periods of time[7] due to the convenience

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of measuring light exposure under a variety of metrics from one device and availability of such devices. Previously, researchers have used ambulatory monitors worn on the wrist to measure the light that individuals may be exposed to daily while undergoing their day-to-day tasks.[1] A study confirmed GENEActiv's (Activinsights, Cambridgeshire, UK) accuracy in mapping light patterns indoors and outdoors when compared to a gold standard light meter (J17 Luma Color, Portland, OR, USA) under LED and fluorescent light in various settings.[8] However, monitors worn on the wrist may produce errors when estimating retinal light exposure due to its positioning.[1] Furthermore, wrist-worn monitors may encounter several issues with capturing light exposure due to factors such as clothing (e.g. especially in cooler climates when long sleeves are worn) and general placement of the arms throughout the day (placed on or under objects) causing differences compared to actual retinal light exposure.[9] Therefore, it has been suggested that a device that measures light exposure closer to the eye is required,[10] and it is also required to measure the validity and reliability of such devices under a wide range of light exposure patterns, for example, indoors and outdoors to establish their efficacy at different light intensities.

Other devices that allow the opportunity to collect light exposure closer to the eye, either by using a monitor that can clip onto clothes or monitors that clip onto spectacles, have recently become available. The MyLyt (LVPEI, Hyderabad, India) is a device that attaches to the participant's clothing (as close to the eye as possible) and uses ambient light sensors to quantify light exposure. It has been validated to differentiate both indoor and outdoor environments and has been reported to show good repeatability, correlation and limited variation in data recording when compared to a digital lux meter in a controlled environmental set-up, and natural outdoor and indoor settings across a day.[11] A glasses-mounted light meter Clouclip Model M2 (HangZhou Glasson Technology, Hangzhou City, China) that is worn next to the user's right temple on a glasses frame using a rubber sleeve has also been developed. This device uses a light intensity sensor to record eye-level ambient illumination every 5 s. This device has been shown to have good and excellent inter-device reliability for illumination and viewing distance measures respectively, but when compared to the gold standard Hagner-S2 photometer (Hagner, Solna, Sweden), it has been reported that the Clouclip underestimates illuminance values when subjected to higher levels of illumination ($>$2500 lx).[12]

A recently released, commercially available device (LYS software, LYS Technologies Ltd™, Copenhagen, Denmark) is a lightweight light sensor that can be clipped onto any part of clothing close to the eye (e.g. shirt collar). Unlike previously mentioned devices, this light meter monitors and tracks the intensity, spectrum and pattern of light exposure under a variety of metrics. It tracks the amount and spectral properties of light that a participant is exposed to every 15 s and transmits data via Bluetooth to a smartphone containing LYS software. The LYS button sensors, or 'light button', measure different light metrics such as photopic illuminance in lux, and also melanopic equivalent daylight illuminance (mEDI), and has been used in free-living conditions,[13] hospital settings,[14] workplace settings[15] and recently, in university students.[16] However, to our knowledge, no studies have investigated the validity of these devices in comparison to a factory-calibrated light meter in both indoor and outdoor environments or researched their inter-device reliability. Therefore, this study aims to measure the validity and reliability of the light buttons against a factory standard digital light meter (DLM) in both indoor and outdoor environments.

2. Method

2.1 Devices

2.1.1 Wearable light sensor

The wearable light sensor, or the 'Light Button' (LYS Button by LYS Technologies Ltd) is a lightweight wearable sensor that collects ambient light in the visible and near-infrared spectra. The sensor utilises red (R), green (G) and blue (B) optical filters to derive correlated colour temperature (in Kelvin), photopic illuminance (measured in lux) and mEDI (also measured in lux). The light button has a reported range of 0 lx to 100 000 lx illuminance, a spectral sensitivity of 380 nm to 1100 nm (R, G, B, IR) and contains a 3-axis accelerometer. The device is approved by both Conformité Européenne (CE) and Federal Communications Commission (FCC). Typically, the light button would be placed on the collar of a human participant – as close to the eyes as feasible.

2.1.2 Digital light meter

The factory-calibrated digital illuminance light meter (RS PRO RS-92 Light Meter, London, UK) or 'DLM' was used as a reference for validating the measurement of photopic illuminance (lux). According to manufacturers, this device has an output range of 0 lx to 40,000 lx, 0.1 lx resolution and ± 3% repeatable accuracy (RS PRO, 2019), with spectral sensitivity between 380 nm and 780 nm. DLMs from this manufacturer have been utilised in previous research.[17]

2.2 Experimental design

2.2.1 Validity

Two light buttons and the DLM (see Figure 1) were tested simultaneously in an indoor environment and an outdoor natural lighting environment on two different occasions in Melbourne, Australia (coordinates = 37.8136°S, 144.9631°E) during Australian Eastern Daylight Time. To assess the light exposure under natural lighting conditions, all devices were placed side-by-side outside in full view of the sky from ~0900 h to ~1700 h, for a total duration of 8 h. Natural daylight duration started at 0714 h (sunrise) to 1719 h (sunset). For the indoor data collection, all devices were placed on a table with light buttons facing upwards, approximately 50 cm from a window with unobstructed light from ~0900 h to ~1700 h. The sensors were placed on a desk at a height of ~73 cm. Since the DLM exclusively captures light exposure in lux, only lux values were utilised from the light buttons for analysis, disregarding other spectral properties provided by the light buttons. Data were collected from the two light buttons, synced to an iPad, and transferred to a computer for further analysis. Data collected from the DLM were also synced to an iPad via a dedicated application and transferred to a computer.

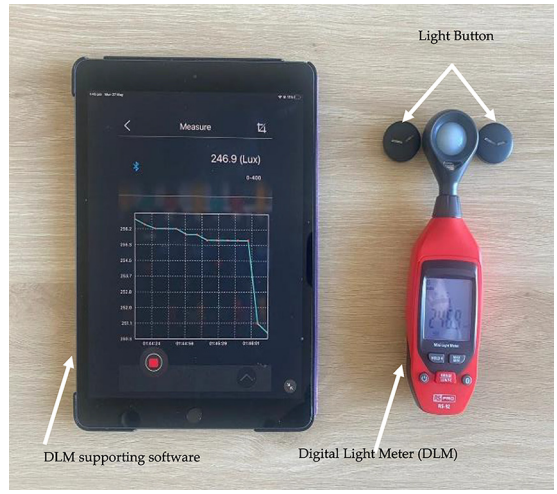


Figure 1 The set-up of the light buttons and DLM on a table alongside an iPad connected to the light buttons

2.1.2 Reliability

The two light buttons were tested simultaneously in the same indoor environment and outdoor natural light environment as the validation procedures on two different occasions in Melbourne, Australia. Both devices were placed side by side from 1500 h to 1700 h, for a total duration of 2 h. Data were collected from both light buttons, synced to an iPad and transferred to a computer for reliability analysis.

2.3 Statistical analysis

Data from devices in both conditions (indoors and outdoors) were exported and converted to 1-min epochs using Python 3.8 (Python Software Foundation, Wilmington, DE, USA). Outlier detection was performed using the interquartile range (IQR) method.[18] Outliers were defined as any data point beyond 1.5 times the IQR above the third quartile or below the first quartile. Any data points outside of the bounds were removed from the analysis as the cause of outliers may have been due to human error from the researcher such as causing shadows over devices, uncontrolled environmental conditions such as sudden change in lighting conditions, or device error throughout the data collection period. Once outliers were removed from analysis, the total number of data points for final validity analysis were 301 in the indoor environment, and 476 for the outdoor environment. Outlier detection for the reliability dataset resulted in 120 data points and 126 data points for the indoor and outdoor environments, respectively. More outliers were found in the indoor condition due to various light intensities being captured from different sources (e.g. artificial light and natural light through a window). Data from both conditions were then transferred into a Microsoft Excel file for validity and reliability analysis.

Descriptive statistics of illuminance values derived from the two light buttons and the DLM are shown as mean ± standard deviation (SD). Validity statistics comparing the lux values derived from the DLM and the light button were calculated using a customised Excel spreadsheet (Hopkins, 2019)[19]. Validity analysis between the light buttons and the DLM was examined by calculating the typical error of estimate (TEE), mean differences and upper and lower limits of agreement (1.96 SD or 95% limits of agreement). The TEE (also known as the standard error of estimate) measures the predictive accuracy of a regression model by indicating how closely the regression line fits the data. The mean difference represents the average difference between paired observations in two related datasets, in this case, lux values derived from the light buttons and the DLM. Pearson's correlation coefficients (r) measure the strength and direction of the linear relationship between the lux values obtained from the light buttons and DLM. Pearson's correlation was calculated with 95% confidence intervals (95% CI) and interpreted using the following thresholds: <0.1, *trivial*; 0.1 to 0.3, *small*; 0.3–0.5, *moderate*; 0.5–0.7, *large*; 0.7–0.9, *very large*; and 0.9–1.0, *almost perfect*.[20] Statistical significance was set at $p < 0.05$. Lastly, the coefficient of variation (CV%) measures the relative variability of data. This was calculated with the associated 95% CI using the following thresholds to interpret results: 10%, *low*; 11%–20%, *acceptable* and >20% used as the cut-off.[21] Two-hour snippets of data were exported from the data collection for reliability analysis of the light buttons in both conditions. Reliability analysis of the light buttons included between-device TEE, mean differences and intraclass correlation

Table 1 Descriptive statistics showing the mean ± SD values for lux values derived from the light button and the DLM in an indoor environment and outdoor environment for ~8 h

Device	Mean ± SD (lx)	TEE (95% CI) (lx)	Mean difference ± SD (lx)	Range of mean difference (lx)	Pearson correlation coefficient (95% CI)	CV% (95% CI)
DLM (indoors)	636.2 ± 248.4	71.1 (66.8 to 76.1)	−24.7 ± 76.2	−31.8 to −17.7	0.96 (0.95 to 0.97)	11.5 (10.8 to 12.4)
Light button (indoors)	611.5 ± 210.4					
DLM (Outdoors)	1543.7 ± 325.6	24.6 (23.1 to 26.3)	82.3 ± 65.2	76.4 to 88.2	1.00 (1.0 to 1.0)	1.8 (1.7 to 1.9)
Light button (Outdoors)	1626.2 ± 385.1					

Validity analysis includes TEE with associated 95% CI, mean difference, range of mean difference (1.96 × SD), Pearson's correlation coefficient with 95% CI and coefficient of variation (CV%) with associated 95% CI for lux values between the light button and DLM.

coefficients (ICCs) with associated 95% CIs and was determined using a customised Excel spreadsheet.[22] Linear regression and Bland–Altman analyses were performed to compare the differences in light exposure levels recorded with the light button and the DLM.

3. Results

Descriptive statistics for lux values derived from the light button and the DLM in each environment are shown in Table 1. Light exposure from both devices over the indoor data collection period and the outdoor data collection period is illustrated in Figure 2.

3.1 Validity

The statistical test results for validation of the light button against the DLM in both an indoor and outdoor environment are shown in Table 1. The mean difference between the two devices was 24.7 ± 76.2 lx in an indoor environment, with the light button reading lower on average than the DLM. The light exposure measurements in an indoor environment resulted in a very strong correlation between both devices. Levels of agreement (Bland–Altman) plots showing ± 95% limits of agreement between the light button and the DLM are displayed in Figure 3. The mean difference between the two devices in an outdoor environment was 82.3 ± 65.2 lx, indicating the light button reading was higher on average compared to the DLM. The light exposure measurements in an outdoor environment resulted in a perfect correlation between the two devices. Levels of agreement (Bland–Altman) plots showing ± 95% limits of agreement between the light button and the DLM in an outdoor environment are also displayed in Figure 3. Linear regression plots comparing the correlation of lux values derived from both devices in an

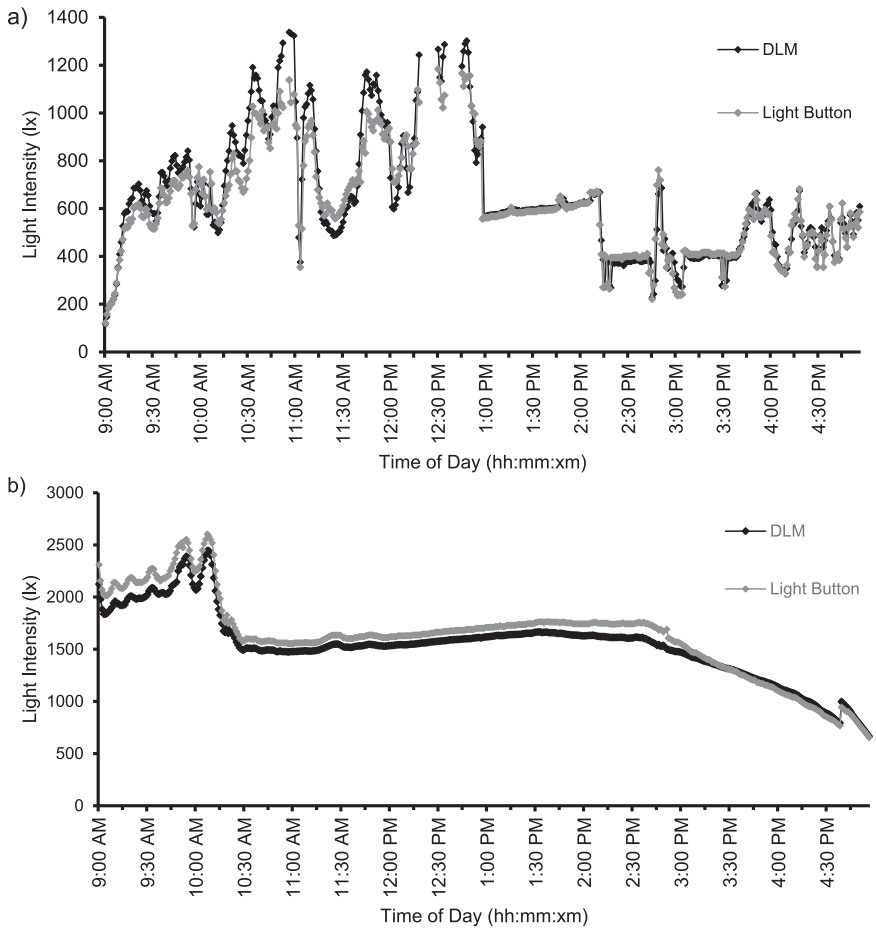


Figure 2 Light intensity (in lux) measured by light buttons (in grey) and the DLM (in black) every 30 min throughout the data collection period in (a) an indoor environment and (b) an outdoor environment

indoor and outdoor condition are displayed in Figure 4.

3.2 Reliability

Reliability analysis of the light buttons in both light environments is shown in Table 2. ICCs between devices were very strong in both indoor and outdoor environments (ICCs = 0.90–0.98) with relatively low TEEs.

4. Discussion

This study aimed to investigate the validity and reliability of wearable light buttons against a factory-calibrated DLM in measuring light exposure in an indoor and outdoor environment. To our knowledge, this is the first study to investigate the validity and reliability of these novel devices. The findings of this study are important, as up until recently, light exposure has been mainly measured using wrist-worn devices,[23]

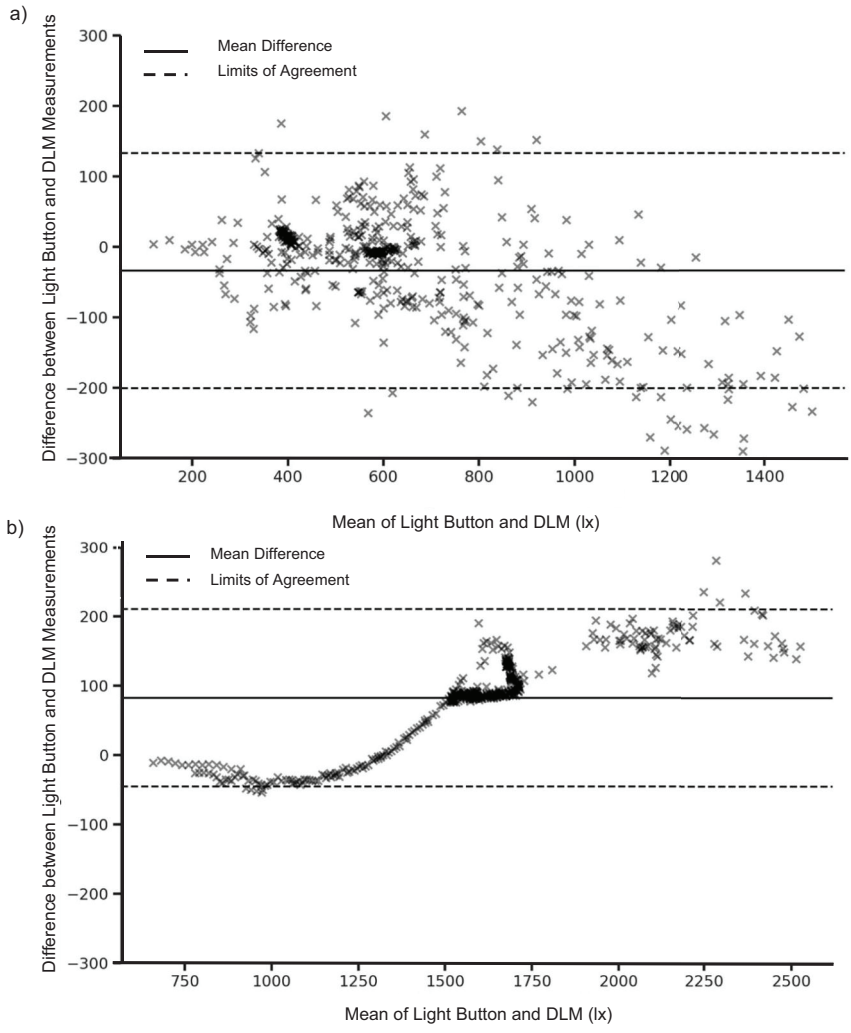


Figure 3 Bland–Altman plot showing the difference between the 'Criterion' (DLM) and 'Practical' (light button) lux measurements against their averages in (a) an indoor lighting environment and (b) an outdoor environment. The middle solid line represents the mean difference, and the two outer dashed lines represent the upper and lower limits of agreement (mean difference ± 1.96 SD)

which may not accurately reflect retinal light exposure. While the wearable light buttons used in the current study are designed to measure light exposure closer to the eye (e.g. on the collar, attached to a hat or potentially mounted on a spectacle frame), data collection occurred in a controlled environment to establish their validity and reliability *in vitro*. It was found that the light button showed very strong correlations, low TEE, mean difference and an acceptable CV%[21] compared to the DLM in an indoor environment. Furthermore, when exposed to light outdoors, a

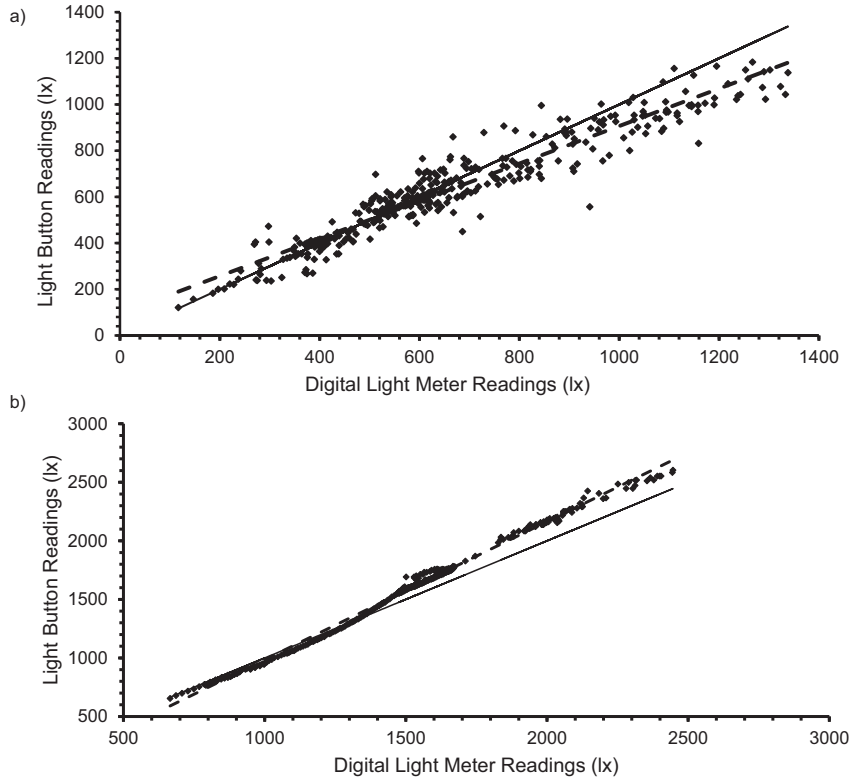


Figure 4 Linear regression (dashed line) between light exposure levels measured by the light button and the DLM in (a) an indoor lighting environment and (b) an outdoor lighting environment. The solid line represents 1:1 relationship

Table 2 Reliability analysis including TEE, mean difference and intraclass correlation coefficients (ICCs) of the lux values derived from the light buttons for ~2 h in both environments

Environment	TEE (95% CI)/lx	Mean difference (95% CI)/lx	ICC (95% CI)
Indoor environment	10.7 (9.5 to 12.3)	67.1 (64.3 to 69.8)	0.90 (0.87 to 0.93)
Outdoor environment	33.5 (29.8 to 38.2)	45.3 (37.0 to 53.7)	0.98 (0.98 to 0.99)

very strong correlation, low TEE, mean difference and a low CV%[21] were found between the two devices.

The light buttons used in this study have been utilised in previous studies over a range of different lighting environments.[11–13,16] In relation to the indoor lighting environment, the results showed that the light buttons underestimated light exposure levels compared to the DLM by ~25 lx. Previous studies validating the utility of wearable sensors in assessing light exposure, such as the Clouclip and Actiwatch, found that

both devices underestimated illumination compared to a referenced light meter, but the correlation between illuminance outputs was almost perfect ($r^2 = 0.99$ and $r^2 = 0.99$ for the Clouclip and Actiwatch, respectively) under different intensities of light. Additionally, mean differences of 430.9 lx and levels of agreement of 79.4 lx were recorded for the Clouclip and the Actiwatch, respectively.[12] Furthermore, the results of this study follow similar trends to other studies comparing wearable devices to a criterion measure. For instance, light intensities measured by GENEActiv device were reported to have a strong linear relationship with photopic illuminance ($r = 0.99$, $p < 0.01$). Unlike the results of the current study, Stone *et al.*[8] reported that an underestimation of light intensity levels was more prominent under lower light intensities with the GENEActiv device. This is of importance due to the human circadian system being more sensitive to lower intensities of light (below 100 lx) compared to high light intensities (up to 2000 lx) in the evening.[24]

The light buttons overestimated light exposure in an outdoor environment compared to the DLM by ~82.5 lx. However, both devices demonstrated a strong correlation, low TEE and mean difference, and a low CV%. Unlike the current study, previous research has shown that some wearable light sensors show increased disparity as light intensity increases.[12] Both the Actiwatch and Clouclip underestimated illumination levels in comparison to a gold standard photometer when exposed to high intensities of light obtained outdoors (>2500 lx), with sensitivity and specificity of 99.7% and 91.7%, respectively.[12] Therefore, the light buttons used in the current study can be utilised in both an indoor and outdoor setting, but further investigation in free-living conditions is required.

Reliability results demonstrated that the light buttons had a high ICC in both indoor and outdoor settings. Previous studies have shown that another wearable light sensor, the MyLyt device, also displayed high ICCs during intra-test reliability testing (ICC = 0.99) in an indoor location ranging from 4 lx to 810 lx.[11] Furthermore, high reliability was found among 10 GENEActiv devices under both fluorescent and LED light (ICC = 0.99 and 0.99, respectively).[8]

The findings of the present study indicate that the TEE between the two light button devices was lower in indoor conditions (~10 lx) but higher in the outdoor condition (~33 lx). Despite this variation, the reliability of the buttons was confirmed for both indoor and outdoor environments. However, future research should consider the TEEs when analysing and interpreting the meaningful differences in light exposure.

We acknowledge several limitations in the current study. Firstly, numerous data points were excluded from analysis due to device errors encountered under varying light intensities in the indoor condition (e.g. negative illuminance values). These errors highlight inherent challenges associated with the device's performance, and we would recommend these error values be filtered out before using the data. Secondly, even though results from the current study investigate the validity and demonstrate the reliability of the light buttons in a controlled setting, there remain gaps in the literature regarding the performance of light buttons in free-living conditions, for example, when worn on the body of humans in daily life. Therefore, the strong correlation observed in this study may not directly translate into the varied environmental conditions from day-to-day life. Furthermore, the testing conditions in the current study do not necessarily reflect the full light intensity range that might be experienced outdoors. The highest lux reading in the outdoor environment was ~2600 lx, which may be considered lower than expected on a

clear day outdoors, where values of up to 100 000 lx may be experienced. Additionally, increasing the number of devices and testing environments/settings over a longer duration of time could enhance the quality of the study, and provide information on the equipment's durability and performance over time. Therefore, future research should investigate the light button performance under a wide variety of conditions, such as higher light intensities and under different light sources, for example, fluorescent or LED light[6,8] while also addressing these confounds.

The present study investigated the validity and reliability of the wearable light buttons against a factory-calibrated DLM in measuring light exposure in lux in both an indoor and outdoor environment. Overall, the findings indicate that the light buttons may serve as valid and reliable tools for monitoring light exposure in both indoor and outdoor controlled environments. However, further research is necessary to validate these devices in measuring the non-visual effects of light as done in previous studies.[25] Another important consideration for future research is the optimal placement of the device, to ensure that there is no shadowing by the head, chin or any clothing. Furthermore, measurements in real-world settings on personal circadian light exposure[26] is required across diverse populations, such as adolescents, athletes and the elderly, to accurately assess light exposure habits and further explore the relationship between light exposure and other circadian health indicators, including sleep and physical activity.

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References

1 Dumont M, Beaulieu C. Light exposure in the natural environment: relevance to mood and sleep disorders. *Sleep Medicine* 2007; 8: 557–565.
2 Borbely AA. A two process model of sleep regulation. *Human Neurobiology* 1982; 1: 195–204.
3 Yin J, Julius AA, Wen JT. Optimization of light exposure and sleep schedule for circadian rhythm entrainment. *PloS One* 2021; 16: e0251478.
4 Walsh NP, Halson SL, Sargent C, *et al*. Sleep and the athlete: narrative review and 2021 expert consensus recommendations. *British Journal of Sports Medicine* 2021; 55: 356–368.
5 Driller MW, Dunican IC, Omond SET, *et al*. Pyjamas, polysomnography and professional athletes: the role of sleep tracking technology in sport. *Sports* 2023; 11: 1–17.
6 Bierman A, Klein TR, Rea MS. The Daysimeter: a device for measuring optical radiation as a stimulus for the human circadian system. *Measurement Science and Technology* 2005; 16: 2292–2299.
7 Martin JS, Gaudreault MM, Perron M, *et al*. Chronotype, light exposure, sleep, and daytime functioning in high school students attending morning or afternoon school shifts: an actigraphic study. *Journal of Biological Rhythms* 2016; 31: 205–217.
8 Stone JE, McGlashan EM, Facer-Childs ER, *et al*. Accuracy of the GENEActiv device for measuring light exposure in sleep and circadian research. *Clocks & Sleep* 2020; 2: 143–152.
9 Scheuermaier K, Laffan AM, Duffy JF. Light exposure patterns in healthy older and young

adults. *Journal of Biological Rhythms* 2010; 25: 113–122.
10 Tähkämö L, Partonen T, Pesonen AK. Systematic review of light exposure impact on human circadian rhythm. *Chronobiology International* 2019; 36: 151–170.
11 Dhakal R, Rudrapankte JR, Chittajallu HSNS, *et al*. Development and validation of a 'MyLyt' wearable light tracking device. *Ophthalmic and Physiological Optics* 2023; 43: 132–140.
12 Howell CM, Mccullough SJ, Doyle L, *et al*. Reliability and validity of the Actiwatch and Clouclip for measuring illumination in real-world conditions. *Ophthalmic and Physiological Optics* 2021; 41: 1048–1059.
13 van Lieshout-Van Dal E, Snaphaan L, Bouwmeester S, *et al*. Testing a single-case experimental design to study dynamic light exposure in people with dementia living at home. *Applied Sciences (Switzerland)* 2021; 11: 10221.
14 Davidoff H, Van den Bulcke L, Vandenbulcke M, *et al*. Toward quantification of agitation in people with dementia using multimodal sensing. *Innovation in Aging* 2022; 6: igac064.
15 Aries MBC, Fischl G, Lowden A, *et al*. The relationship of light exposure to sleep outcomes among office workers. Part 1: Working in the office versus at home before and during the COVID-pandemic. *Lighting Research and Technology* 2024; 56: 113–125.
16 Khanie MS, Illum T, Hosseinpour N, *et al*. Exploring the effects of spectral light exposure on University students' sleep quality: a cross-sectional study. *E3S Web of Conferences* 2023; 396: 1–6.
17 Macleod H, Weiss L, Kelliher S, *et al*. The effect of UVA light/8-methoxypsoralen exposure used in Extracorporeal Photopheresis treatment on platelets and extracellular vesicles. *PLoS One* 2024; 19: e0293687.
18 Rousseeuw PJ, Hubert M. Anomaly detection by robust statistics. *Wiley Interdisciplinary Reviews Data Mining Knowledge Discovery* 2018; 8: e1236.
19 Hopkins WG. Spreadsheets for analysis of validity and reliability. *Sportscience* 2019; 21.
20 Atkinson G, Nevill A. Measures of reliability in sports medicine and science. *Sports Medicine* 2000; 30: 375–381.
21 Shechtman O. *The coefficient of variation as an index of measurement reliability.* Methods of clinical epidemiology. BHSBH 2013. 39-49. Epidemiology and Public Health Methods of Clinical Epidemiology. 2013.
22 Hopkins WG. Spreadsheets for analysis of validity and reliability. *Sportscience* 2015; 19: 36–44.
23 Khodasevich D, Tsui S, Keung D, *et al*. Characterizing the modern light environment and its influence on circadian rhythms. *Proceedings of the Royal Society B: Biological Sciences* 2021; 288: 20210721.
24 Phillips AJK, Vidafar P, Burns AC, *et al*. High sensitivity and interindividual variability in the response of the human circadian system to evening light. *Proceedings of the National Academy of Sciences* 2019; 116: 12019–12024.
25 Stampfli JR, Schrader B, di Battista C, *et al*. The Light-Dosimeter: A new device to help advance research on the non-visual responses to light. *Lighting Research and Technology* 2023; 55: 474–486.
26 Figueiro MG, Hamner R, Bierman A, *et al*. Comparisons of three practical field devices used to measure personal light exposures and activity levels. *Lighting Research and Technology* 2013; 45: 421–434.